

Convergence of $\sum (-1)^n \frac{\sin^2(n)}{n}$

The solution provided in class was incorrect; here is the revised and complete solution.

Note that

$$\sum_{n=1}^{\infty} \frac{(-1)^n \sin^2(n)}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} - \sum_{n=1}^{\infty} \frac{(-1)^n \cos(2n)}{2n}$$

which converges since

- $\sum \frac{(-1)^n}{2n}$ converges
- $\sum \frac{(-1)^n \cos(2n)}{2n}$ converges

indeed by **Abel transformation** and **Lagrange's trigonometric identities**, let

$$a_n = \frac{(-1)^n}{2n}, \quad b_n = \cos(2n) = B_n - B_{n-1}, \quad B_n = \sum_{k=1}^n \cos(2k) = -\frac{1}{2} + \frac{\sin(2n+1)}{2 \sin 1}.$$

$$\begin{aligned} S_N &= \sum_{n=1}^N a_n b_n = \sum_{n=1}^N \frac{(-1)^n \cos(2n)}{2n} = \frac{(-1)^N}{2N} \left(-\frac{1}{2} + \frac{\sin(2N+1)}{2 \sin 1} \right) - \\ &\quad \sum_{n=1}^{N-1} \left[\frac{1}{2} + \frac{\sin(2n+1)}{2 \sin 1} \right] \left(\frac{(-1)^n}{2n+2} - \frac{(-1)^n}{2n} \right) = \\ &= \frac{(-1)^N}{2N} \left(-\frac{1}{2} + \frac{\sin(2N+1)}{2 \sin 1} \right) + \sum_{n=1}^{N-1} \frac{(-1)^n (2n+1)}{2n(2n+1)} - \sum_{n=1}^{N-1} \frac{(-1)^n \sin(2n+1)}{4n(n+1) \sin 1}. \end{aligned}$$

and

$$\sum_{n=1}^{\infty} \frac{(-1)^n \sin(2n+1)}{4n(n+1) \sin 1}$$

converges absolutely by comparison with $\sum \frac{1}{n^2}$.